Reanalysis of Dugstad, Hoel, and Lindhjem (2026): "Kraft med mindre konflikt? Preferanser for sol- vs. vindkraft"

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Abstract

This mini-report summarizes the re-analysis of the solar-versus-wind discrete choice experiment using the same data as Dugstad, Hoel and Lindhjem (2026) and with an independent R implementation. It verifies the published calculations, replicates the benchmark mixed logit, examines observed and unobserved preference heterogeneity, estimates latent-class and hybrid-choice models, formally compares the technologies, and reassesses policy conclusions using cost-per-kWh and acceptance curves. Our evidence reproduces the main qualitative ranking of solar and wind, and shows that the status quo component is sensitive to modelling and exclusion rules, while the marginal willingness-to-accept estimates for no protection, proximity and visibility keep their sign and order of magnitude across specifications. A pooled mixed-logit block test is incomplete because several first-round models stopped at sub-optimal likelihood values; those results are therefore not used here.

1 Data and baseline framework

The data, which I fetched from Anders' GitHub repository (link at the end of the report), comprises 775 respondents evaluating solar power and 788 evaluating wind power. Each respondent completed eight choice tasks, yielding 6,200 solar and 6,304 wind observations. Each task involved choosing between a "development alternative" and the "status quo" of no development. Development alternatives varied in electricity production, number of plants, protection of nature, distance from residences, landscape visibility, and household's annual electricity bill reduction. The bill reduction, offered annually for 10 years, serves as the compensation or monetary scale for the direct estimation of utility in willingness-to-accept (WTA) space.

The choice architecture is embedded within a Random Utility Maximisation (RUM) framework where respondent n evaluates a choice set $J_{nt} = \{\text{Development Alternative } (A),\ \text{Status Quo } (SQ)\}$ across tasks $t = 1, \ldots, 8$. To express valuations directly in monetary terms, utility is parameterized in Willingness-to-Accept (WTA) space:

$$U_{nit} = \begin{cases} \sigma_n \left(B_{nit} - \sum_{k=1}^{K} w_{nk} X_{nkit} \right) + \varepsilon_{nit} & \text{for } i = A \\ \sigma_n \, w_n^{SQ} + \varepsilon_{nit}^{SQ} & \text{for } i = SQ \end{cases} \tag{1}$$

where B_{nit} is the annual household electricity bill reduction (offered for 10 years), $\sigma_n > 0$ is the cost-scale parameter (the marginal utility of income), X_{nkit} are the K non-monetary project attributes, w_{nk} is the willingness to accept for moving attribute k away from its reference level, and w_n^{SQ} is the willingness to accept for giving up the status quo, that is, the status quo constant expressed in money. The corresponding preference-space coefficients are $\beta_{nk} = -\sigma_n w_{nk}$ and $\alpha_n^{SQ} = \sigma_n w_n^{SQ}$. Assuming the unobserved stochastic components ε_{nit} are independently and identically distributed Type I

Extreme Values, choice probabilities follow the logit form, and every w is directly interpretable in NOK per household per year.

The data includes socio-demographic characteristics, including: exposure to energy development, organisational membership or associations, attitudes, perceived consequentiality, response time, and choice difficulty. Respondent identifiers account for the repeated-choice panel structure. Persistent status quo choice is substantial, however, because 35.9% of solar respondents and 52.0% of wind respondents selected "no development" in all eight tasks presented. I retain these respondents in the main sample and examine them through latent-class and robustness analyses rather than automatically classifying them as protesters. The age distribution closely matches the adult population of Innlandet county, but education is strongly skewed. Only 5.3% of the respondents in solar sample and 4.7% of the respondents in wind sample have basic schooling, compared with 26.1% in the population. Raking weights correct age-by-sex and education margins, but reduce the effective sample size to approximately 398 for solar and 365 for wind. The weighting has little effect on the main status quo shares.

2 Published estimates and our benchmark replication

The published Figure 6 is reproduced exactly at the reported precision from Table A.3. The updated check also identifies two inconsistencies. First, the wind worked-out example in Table A.6 uses the 57,000-household coefficient and gives 17.16 øre/kWh. Figure 6 uses the 38,000-household coefficient and correctly gives 17.62 øre/kWh; the inconsistency is in the worked-out example, not Figure 6. Second, the solar status quo value is reported as NOK 3,822 in the text but NOK 3,872 in Table A.3.

The independent mixed-logit replication uses the published WTA-space specification econometrically, estimates 30 parameters, with 500 MLHS draws rather than the paper's 2,000 draws. Both replications converged successfully. The solar log-likelihood improves from −2536.4 to −2531.3, and the wind log-likelihood from −2065.2 to −2054.7. The corresponding BIC values are 5,324.6 for solar and 4,371.9 for wind, compared with the published values of 5,335 and 4,392. The qualitative conclusions are reproduced, although several weakly identified marginal means, including some production and plant-number effects, differ.

Table 1: Benchmark mixed-logit replication

	Solar		Wind	
	Published	Replication	Published	Replication
Log-likelihood	−2536.4	−2531.3	−2065.2	−2054.7
BIC	5,335	5,324.6	4,392	4,371.9
Parameters	30	30	30	30

The published wind standard errors are much smaller than those from the replication. For example, the status quo mean has a published standard error of 0.703, compared with a robust replication standard error of 5.70, both in units of 100 NOK. But because the discrepancy persists in our updated analysis, we need to cross-check these findings before we make any strong precision claims.

3 WTA and marginal WTA results

Table 2 compares the main WTA results. The status quo term is an alternative-specific constant capturing unobserved factors favouring no development; it is not a pure nature valuation. The other rows are marginal WTA values for attribute changes relative to their reference levels.

The broad pattern is stable: lack of protection and proximity require substantial compensation, and

Table 2: Main WTA and marginal WTA estimates (NOK per household-year)

	Solar			Wind		
Component	Paper	Replication	Full covariance	Paper	Replication	Full covariance
Status quo	3,872	4,059	2,189 + shifters	13,726	14,661	13,683 + shifters
No protection	7,429	6,643	10,754	5,284	7,440	5,555
Distance below 1 km	4,359	5,038	2,968	9,551	9,967	10,033
Visible	4,699	3,858	6,586	2,800	2,378	3,174

Note: Shifters are respondent characteristics (age, sex, education, income, exposure, organisation membership) that shift the status quo term.

wind is more sensitive to distance. However, weaker production and plant-number effects vary across specifications. The full-covariance status quo estimates refer to respondents at the reference values of the shifters and therefore are not directly equivalent to the simple constants in the paper and replication.

4 Observed heterogeneity and full-covariance mixed logit

In the baseline MNL , adding exogenous status quo shifters improves fit substantially(likelihood-ratio $p < 10^{-22}$ for both technologies). In the fullest MNL specification, which also controls for attitudes and the belief that the results will be used,, age increases the status quo WTA by approximately NOK 1,359 per decade for solar ($t = 3.5$) and NOK 1,369 for wind ($t = 3.0$). Women have higher status quo WTA by approximately NOK 2,258 for solar ($t = 1.8$) and NOK 3,307 for wind ($t = 2.2$); the solar effect is weak. Education and income show no systematic effects ($|t| \leq 0.8$ for both technologies), and cost sensitivity does not vary significantly with income ($t = 1.0$ for solar and 1.1 for wind). The effect of exposure is not significant in this rung ($t = -1.3$ and -1.1) because the attitude-related variables absorb it; the hybrid model in Section 7 shows exposure effects through the latent attitude. Attitudes produce much larger fit improvements, but they are potentially endogenous and should therefore be interpreted carefully.

A five-dimensional full-covariance mixed logit was estimated in stages for the status quo, no protection, distance below 1 km, visibility, and natural logarithm of cost. The model converges for both technologies. Relative to the published model, the BIC improves from 5,335 to 5,320 for solar and from 4,392 to 4,388 for wind. But relative to our own replication, it is almost unchanged for solar (5,325 versus 5,320) and worse for wind (4,372 versus 4,388). The richer covariance structure therefore improves the fit compared to the published model but not to our own replication for wind. The random-cost dimension significantly improves likelihoods relative to the fixed-cost stage, although it does not improve BIC for wind relative to that stage. Large correlations appear among several taste dimensions, whereas correlations between tastes and the scale dimension remain modest. These results indicate that the papers's restricted covariance structure may omit important preference relationships, but the unusually large t-ratios in the random-coefficient models remain a concern.

5 Latent classes and non-compensatory behaviour

The three-class model distinguishes cost-responsive traders, respondents who respond to the project attributes but ignore compensation, and respondents who almost always select the status quo. The trader cost coefficient is fixed at its two-class value because the unrestricted wind model collapsed toward a second cost-ignorer class.

Formally, the prior probability that respondent n falls into a specific latent class $c \in \{1, 2, 3\}$ is evaluated via a multinomial logit system governed by individual covariates and indicators of stated

cost importance (Z_n):

$$P_{nc} = \frac{\exp(\boldsymbol{\gamma}_c \boldsymbol{Z}_n)}{\sum_{j=1}^{3} \exp(\boldsymbol{\gamma}_j \boldsymbol{Z}_n)} \tag{2}$$

with $\boldsymbol{\gamma}_3$ restricted to a zero vector for identification purposes. Conditional on class membership, the panel probability of a choice sequence is driven by class-specific taste parameters. The non-compensatory behaviors detailed in our empirical taxonomy are isolated by applying strict structural linear constraints on the class utility functions:

- **Class 1: Cost-Responsive Traders ($c = 1$):** These households evaluate alternative balances continuously, tracking parameters where cost sensitivity $\sigma_1 > 0$ is fixed to the baseline two-class solution ($\sigma_1 = \bar{\sigma}$), and standard attribute tastes are active ($\beta_1 \neq 0$, $\alpha_1^{SQ} \in R$).
- **Class 2: Cost-Ignorers ($c = 2$):** These respondents display non-compensatory choice behavior by evaluating technical project characteristics while tuning out financial dimensions completely, operationalized by fixing cost sensitivity to zero:

$$\beta_2 \neq 0, \quad \sigma_2 \equiv 0, \quad \alpha_2^{SQ} \in R \tag{3}$$

- **Class 3: Status Quo Insisters ($c = 3$):** These respondents almost always choose the status quo. Attributes and cost do not enter their utility ($\beta_3 \equiv 0$, $\sigma_3 \equiv 0$); only an estimated constant α_3^{SQ} remains. Their conditional probability of choosing the status quo is the same in every task and close to, but not exactly, one:

$$P_{n,SQ,t|c=3} = \frac{\exp(\alpha_3^{SQ})}{1 + \exp(\alpha_3^{SQ})} \approx 0.99 \quad \forall\, t \tag{4}$$

  (0.991 for solar and 0.988 for wind).

The unconditional individual sequence probability is subsequently maximized over the sum of these discrete fractions: $P_n = \sum_{c=1}^{3} P_{nc} \prod_t P_{nit|c}$.

Table 3: Estimated latent-class shares

	Solar			Wind		
	Trader	Cost-ignorer	Status quo	Trader	Cost-ignorer	Status quo
Unweighted	25.4%	36.9%	37.8%	15.3%	27.9%	56.8%
Population-weighted	26.5%	36.5%	37.0%	16.5%	28.8%	54.6%

The restricted three-class model has BIC 4,989.5 for solar and 3,881.7 for wind, compared with 5,553.4 and 4,349.1 for the two-class model with attitudes. The cost-ignorer interpretation receives supporting evidence from stated cost importance: in the solar sample, each one-point increase in stated cost importance reduces the odds of membership by about 16% (odds ratio 0.84, $z = -2.3$). The corresponding wind relationship is weaker (odds ratio 0.85, $z = -1.6$). Because the trader classes are small and their WTA estimates are imprecise, class-specific WTA values should not support strong policy conclusions.

6 Formal comparison of solar and wind

Two completed pooled comparisons are reported: a pooled two-class latent-class test and a pooled MNL scale test. In the pooled two-class model, technology-specific constants improve fit strongly

(LR 45.9, 2 df, $p = 1.1 \times 10^{-10}$). Technology-specific covariate effects in class membership are not supported ($p = 0.37$). Allowing the ten trader attribute WTAs to differ gives a moderate likelihood-ratio result (LR 21.7, 10 df, $p = 0.016$), while BIC favours the parsimonious model with technology-specific constants.

The pooled MNL gives a clearer decomposition. Equality of all eleven WTA parameters is rejected when scale may differ (LR 43.4, 11 df, $p = 9.2 \times 10^{-6}$). Once the status quo and scale are free, equality of the other ten marginal WTAs is not rejected (LR 13.7, 10 df, $p = 0.19$). The status quo term differs strongly (LR 29.7, $p = 5.1 \times 10^{-8}$), whereas the remaining scale difference does not (LR 0.7, $p = 0.41$). Thus, in the pooled MNL, the main solar-wind difference lies in the status quo constant rather than the ten marginal attribute valuations.

Table 4: Likelihood-ratio tests of equality between solar and wind

Test	LR	df	p-value
Pooled MNL			
All eleven WTAs equal (scale free)	43.4	11	9.2×10^{-6}
Status quo term equal (scale free)	29.7	1	5.1×10^{-8}
Other ten WTAs equal (status quo, scale free)	13.7	10	0.19
Scale equal	0.7	1	0.41
Pooled two-class latent class			
Technology-specific constants	45.9	2	1.1×10^{-10}
Ten trader attribute WTAs equal	21.7	10	0.016

The pooled mixed-logit block tests are not yet valid. In the first estimation round, the model with equal shifters reached a higher log-likelihood than the unrestricted model containing it. This violates nesting and shows that the unrestricted, scale-restricted, attribute-mean, and covariance models stopped at inferior likelihood values. The restarted block tests are therefore pending and no p-values from the first round are reported. I will present the completed tests at the oral examination.

7 Hybrid choice model

The hybrid choice model extends standard framework configurations by explicitly modeling choice dependencies alongside unobserved subjective frameworks. The underlying structural system maps a latent attitudinal construct (LV_n^*, representing generalized environmental concern) driven by a vector of exogenous characteristics and organizational affiliations (X_n):

$$LV_n^* = \Gamma_{LV} X_n + \zeta_n \tag{5}$$

where ζ_n represents a normally distributed structural disturbance. This latent construct simultaneously feeds into the measurement equations of the five ordered indicators and enters the choice pathway as an explicit utility modifier. Specifically, the alternative-specific constant for the status quo varies systematically according to the structural shift: $\text{WTA}_n^{SQ} = \delta_0 + \kappa_{LV} LV_n^*$. Because the indicator metrics are integrated alongside choice outcomes inside a joint log-likelihood expression,

$$L_{\text{hybrid}} = \prod_n \int_{LV} L_{\text{choice}} \cdot L_{\text{measurement}} \cdot f(LV) dLV \tag{6}$$

it cannot be directly checked against choice-only specifications via standard BIC bounds.

The empirical estimation represents concern about nature with five ordered indicators. All five load strongly on a common latent construct. Higher latent concern increases the status quo WTA.

Exposure to possible energy development raises the status quo WTA through the latent construct by approximately NOK 5,891 for solar and NOK 14,927 for wind. Environmental-organisation membership raises it by approximately NOK 8,290 and NOK 8,586, respectively. Age also has a clear positive effect in the wind sample, while education and income do not.

The hybrid likelihood includes the attitudinal indicators and therefore cannot be compared directly with choice-only model likelihoods or BIC values. The results show associations between exogenous characteristics, latent concern, and choices, but do not establish causal effects because the attitudes may partly rationalise earlier choices.

8 Policy results: cost per kWh and acceptance

The cost-per-kWh estimates of the paper are reproduced exactly at the printed precision. Across the published model, baseline MNL, mixed-logit replication, and full-covariance model, wind has a higher cost per kWh than solar for the reported scenarios. For scenario P6 without visibility, the published values are 3.3 ore/kWh for solar and 9.2 for wind; the mixed-logit replication gives 3.6 and 9.4, the baseline MNL 2.7 and 9.2, and the full-covariance model 3.9 and 10.1.

Acceptance curves reveal information that an average WTA does not. Under the exogenous two-class model, acceptance approaches an upper bound because a large status-quo-only class remains unlikely to accept even at high compensation. At NOK 9,000 per year, population-weighted acceptance for the cheapest scenario (P6) is approximately 47.0% for solar and 35.5% for wind. At NOK 50,000 it reaches only 50.0% and 39.2%. Population weighting changes the acceptance shares by less than one percentage point.

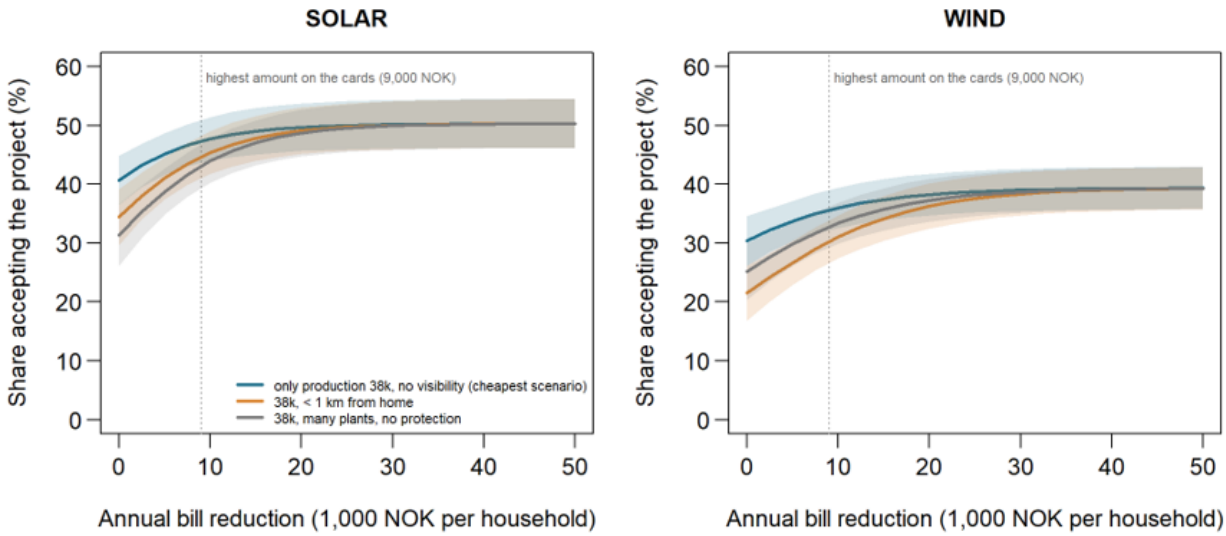


Figure 1: Population-weighted predicted acceptance of three project scenarios as annual household bill reductions increase. The vertical dotted line marks NOK 9,000, the highest compensation amount shown in the choice cards. Amounts above NOK 9,000 are outside the range shown to respondents and are extrapolations. Shaded bands show uncertainty around the predicted acceptance curves.

These results qualify the interpretation of average cost per kWh. Compensation can shift choices among responsive households, but it cannot represent a meaningful average trade-off for respondents whose choices show little or no compensation response.

9 Robustness to exclusions

The marginal WTAs for no protection, distance below 1 km, and visibility are basically stable across alternative respondent screens. However, the status quo component is not stable. In the simple MNL (not the mixed logit of Table 2) the wind status quo term is NOK 13,084 including all respondents, NOK 3,274 after the paper's protest rule, NOK 7,439 when protected-value reasons are not treated as protest, and −6,945 after removing all always-status-quo respondents. The corresponding solar values are NOK 2,749, −3,644, −1,706 and −8,884. These numbers are not comparable with the status quo terms in Table 2, which come from the mixed logit through the sample of the published paper.

The papers's protest rule removes 21.4% of the solar sample and 29.6% of the wind sample. Excluding the protected-value reason reduces these figures to 14.5% and 17.3%. The protected-value statement accounts for 19.4% of always-status-quo respondents in solar and 23.7% in wind. Because this statement expresses a substantive value rather than rejection of the survey, removing them will alter the estimated status quo cost and risks by excluding genuine preferences.

10 Model comparison and evidence limitations

Table 5: BIC comparison for choice models

Model	Solar	Wind
MNL, no heterogeneity	8,237	7,498
MNL + exogenous characteristics	8,174	7,373
MNL + attitudes	6,080	4,710
Published mixed logit	5,335	4,392
Replicated mixed logit	5,325	4,372
Full-covariance mixed logit	5,320	4,388
LC2, constant shares	5,887	4,825
LC2 + exogenous class membership	5,924	4,846
LC2 + attitudes	5,553	4,349
LC3, restricted cost-responsive traders	4,990	3,882

BIC comparisons apply only to choice models estimated on the same data. I excluded the hybrid model because its likelihood also contains indicator responses. The lower BIC of attitude-based latent-class models does not show that attitudes cause choices. The full Bayesian 12-by-12 covariance model did not converge: several Geweke statistics remain far outside ±2 and many posterior correlations approach ±1. Its estimates are therefore exploratory and are not used in our conclusions. The random-coefficient models used 250 to 500 simulation draws (against 2,000 MLHS in published paper); their standard errors, like the published ones, are probably too small, and I did not bootstrap them, as well. The trader classes of the latent-class models are small but their WTAs are imprecise.

11 Conclusion

Subject to verification by rerunning the code, this documented re-analysis supports five conclusions. First, the published papers's main qualitative results and cost-per-kWh ranking are reproducible. Second, no protection, proximity, and visibility have the most stable marginal WTA estimates, while smaller production and plant-number effects are less secure. Third, the general status quo component is the main solar-wind difference in the completed pooled MNL and is highly sensitive

to protest classification and sample exclusion. Fourth, a substantial share of respondents seem not to trade-off project impacts against compensation, so class shares and acceptance curves are more informative than a single population-average WTA. Fifth, exposure and environmental concern are closely related to status quo choices, especially for wind.

The evidence does not support any inference from the unconverged Bayesian model or from the pooled mixed-logit block tests.Those tests are still running on my computer when I prepared this report. I will present them at the oral examination, together with the R code. This analysis therefore relies on the completed MNL and latent-class technology tests. It reports WTA and acceptance results together, shows sensitivity to exclusion rules, and distinguishes marginal attribute values from the status quo components.

References

[1] Dugstad, A., Hoel, J. & Lindhjem, H. (2026). Kraft med mindre konflikt? Preferanser for sol- vs. vindkraft. *Samfunnsøkonomen*, 138(2). https://www.samfunnsokonomen.no/artikkel/kraft-med-mindre-konflikt-preferanser-for-sol-vs-vindkraft/

Link to the data and replication codes to Anders' GitHub repository:
https://github.com/dugstad/Solar-vs-Wind-Experiment